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Unit 1 - Introduction to IoT

PDF Documents

PDF Document 1: 1738214012379-Chapter 6 - Unit 1 - Introduction to IoT.pdf

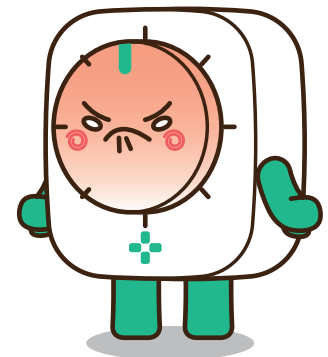
This PDF document is included in the unit materials.



UNIT 1 INTRODUCTION TO IOT



The Internet of Things (IoT) refers to the network of physical objects that are embedded with sensors, software, and other technologies with the purpose of connecting and exchanging data with other devices and systems over the internet. These objects, often called "smart" devices, can range from ordinary household items like refrigerators and lighting fixtures to sophisticated industrial tools.

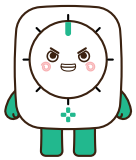




Here are some key aspects of IoT:

- 1 Connectivity
- 2 Sensors and Actuators
- 3 Automation and Control
- 4 Efficiency and Optimization
- 5 Scalability
- 6 Data Processing and Analysis

A smartwatch that tracks your steps and sleep, then sends this data to your phone.

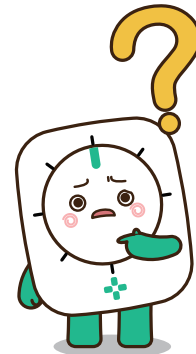


HOW IOT WORKS



Connectivity

IoT devices like smartwatches or home assistants use Wi-Fi or Bluetooth to connect to the internet. This is similar to how a laptop or smartphone connects to the internet to access websites or send messages.



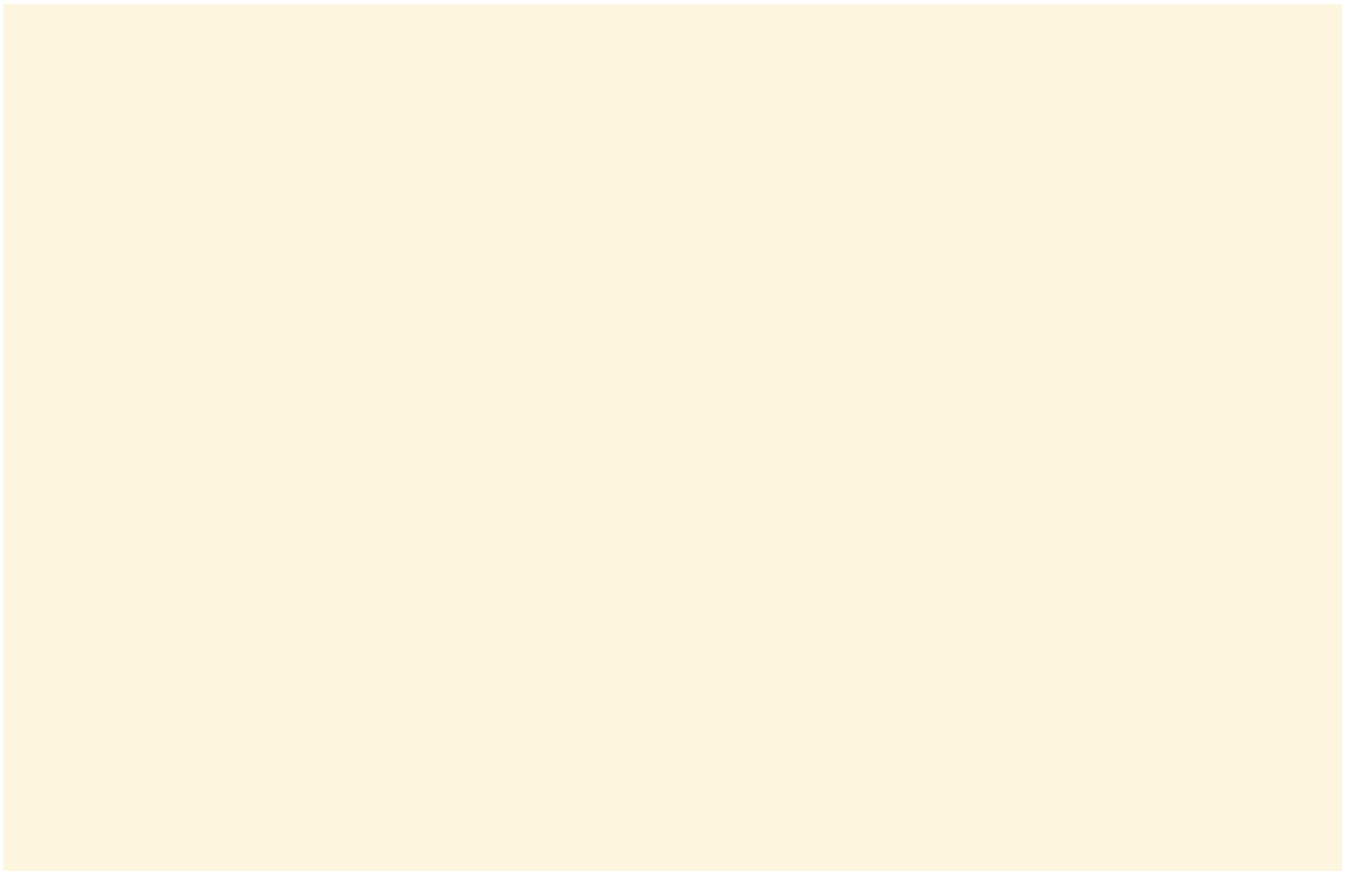


Data Exchange

These devices have sensors that collect information, like temperature or how many steps you take. They then send this information over the internet or to other devices, helping them understand what's happening around you or give you useful feedback.



Draw a diagram showing how a smartwatch sends health data to a phone



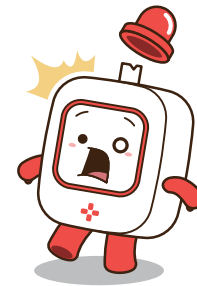


SAFETY AND PRIVACY WITH IOT

What does "being hacked" mean?

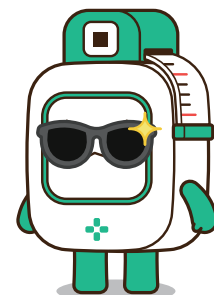


"Being hacked" means someone unwanted finds a way to get into your smart device, like your tablet, smartwatch, or computer, without your permission. This is similar to someone sneaking into your room and taking a look at your secret diary or playing with your toys without asking you.



- 1 How can IoT devices be hacked?
 - A Weak Passwords.
 - B Not Updating
 - C Too Many Connections

- 2 What can happen if an IoT device is hacked?
 - A Privacy Loss
 - B Hackers Control Devices
 - C Spreading the Hack to other devices



●● Activity: Interactive Discussion: _____

Let's name devices they use at home and guess if they can be connected to the internet.





Activity: Safety Tips Poster

Create posters on tips for keeping personal information safe while using smart devices.

